

INFS4203/7203 Data Mining

Tutorial Week 2 - Introduction to Classification

Semester 2, 2025

Question 1/4: Imputation of missing feature values and feature value normalization

Consider a dataset in Table 1.

1. Perform missing value imputation by all-class and class-specific values!
2. Perform max-min feature normalization!
3. Perform z-score feature normalization!
4. (Extra) Does it make sense to apply normalization on categorical (nominal) features? Justify your opinions!

Table 1: A dataset of 6 data points. The right most column is the target classification label.

Index	Age	Home Distance	Browser	Caffeine Level	Hair Style
1	17	100	Chrome	High	Long
2	20	50	Chrome	Medium	Long
3	null	750	Firefox	High	Long
4	27	1000	Chrome	Low	Short
5	26	9500	Safari	Low	Short
6	24	7000	Firefox	null	Short

Answer:

For Age

Numerical: Fill with the average value of the feature.

- All-class: $(17 + 20 + 27 + 26 + 24)/5 = 22.8$
- Class-specific: $(17 + 20)/2 = 18.5$

For Caffeine Level

Nominal: Fill with the most common feature value

- All-class: either High or Low
- Class-specific: Low

Max-min normalization

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (1)$$

- Age: max = 27, min = 17

$$\begin{aligned}(17 - 17)/(27 - 17) &= 0 \\(20 - 17)/(27 - 17) &= 0.3 \\(18.5 - 17)/(27 - 17) &= 0.15 \\(27 - 17)/(27 - 17) &= 1 \\(26 - 17)/(27 - 17) &= 0.9 \\(24 - 17)/(27 - 17) &= 0.7\end{aligned}$$

- Home Distance: max = 9500, min = 50

$$\begin{aligned}(100 - 50)/(9500 - 50) &= 0.0053 \\(50 - 50)/(9500 - 50) &= 0 \\(750 - 50)/(9500 - 50) &= 0.0741 \\(1000 - 50)/(9500 - 50) &= 0.1005 \\(9500 - 50)/(9500 - 50) &= 1 \\(7000 - 50)/(9500 - 50) &= 0.7354\end{aligned}$$

Z-Score normalization:

$$x' = \frac{x - \mu}{\sigma} \quad (2)$$

- Mean:

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i \quad (3)$$

$$\mu = \frac{17 + 20 + 18.5 + 27 + 26 + 24}{6} = \frac{132.5}{6} \approx 22.08$$

- Standard deviation:

$$\sigma = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \mu)^2} \quad (4)$$

$$\sigma = \sqrt{\frac{(17 - 22.08)^2 + (20 - 22.08)^2 + (18.5 - 22.08)^2 + (27 - 22.08)^2 + (26 - 22.08)^2 + (24 - 22.08)^2}{6 - 1}} = \sqrt{17.24168} \approx 4.15$$

- Calculate each feature value:

$$\begin{aligned}(17 - 22.08)/4.15 &= -1.22 \\(20 - 22.08)/4.15 &= -0.50 \\(18.5 - 22.08)/4.15 &= -0.86 \\(27 - 22.08)/4.15 &= 1.19 \\(26 - 22.08)/4.15 &= 0.95 \\(24 - 22.08)/4.15 &= 0.46\end{aligned}$$

It does not make sense, because there are non-numeric variables representing categories without any inherent order.

- Mathematically meaningless
- Implies ordinal relationships where none exist

Question 2/4: Classifier evaluation: accuracy and F1-score on balanced datasets

Consider the following true (ground-truth) class labels y and predicted class labels $\hat{y}_a$ and $\hat{y}_b$ from classifiers A and B, respectively, for 6 data points (instances).

$$y = [1, 1, 1, 0, 0, 0],$$

$$\hat{y}_a = [0, 1, 0, 0, 0, 1],$$

$$\hat{y}_b = [0, 0, 1, 1, 1, 0].$$

1. Calculate the accuracy and F1-score of Classifier A!
2. Calculate the accuracy and F1-score of Classifier B!
3. Which classifier is the best based on their accuracy scores?
4. Which classifier is the best based on their F1-scores?

Answer:

For Classifier A

	1	0
1	TP: 1	FN: 2
0	FP: 1	TN: 2

$$\begin{aligned} \text{Accuracy} &= \frac{TP + TN}{n} = \frac{1 + 2}{6} = \frac{1}{2} \\ \text{Precision} &= \frac{TP}{TP + FP} = \frac{1}{1 + 1} = \frac{1}{2} \\ \text{Recall} &= \frac{TP}{TP + FN} = \frac{1}{1 + 2} = \frac{1}{3} \\ \text{F1} &= \frac{2}{\frac{1}{\text{Precision}} + \frac{1}{\text{Recall}}} = \frac{\frac{1}{3}}{\frac{1}{2} + \frac{1}{3}} = \frac{2}{5} \end{aligned}$$

For Classifier B

	1	0
1	TP: 1	FN: 2
0	FP: 2	TN: 1

$$\begin{aligned}\text{Accuracy} &= \frac{TP + TN}{n} = \frac{1 + 1}{6} = \frac{1}{3} \\ \text{Precision} &= \frac{TP}{TP + FP} = \frac{1}{1 + 2} = \frac{1}{3} \\ \text{Recall} &= \frac{TP}{TP + FN} = \frac{1}{1 + 2} = \frac{1}{3} \\ \text{F1} &= \frac{\frac{2}{3}}{\frac{2}{3}} = \frac{1}{3}\end{aligned}$$

A, because $\frac{1}{2} > \frac{1}{3}$

A, $\frac{2}{5} > \frac{1}{3}$

Question 3/4: Classifier evaluation: accuracy and F1-score on imbalanced datasets

1. Consider a cancer dataset, where 85 people have cancer and the remaining 201 do not. Classifier α always predicts that someone has cancer. Classifier β always predicts that someone does not have cancer. Compare these two degenerated classifiers α and β in terms of their accuracy and F1-scores! Which classifier is better (more preferable) for this domain?

2. Consider a city of 1 million people, where there are 10 COVID-positive individuals. A classifier predicts that there are 1000 COVID-positive individuals (including 10 individuals who are indeed COVID-positive), and the remaining 999,000 are COVID-negative. Analyze this classifier in terms of its accuracy and F1-score!

Answer:

Classifier α

	cancer	no cancer
cancer	TP: 85	FN: 0
no cancer	FP: 201	TN: 0

$$\text{accuracy} = \frac{85}{286} \approx 0.29$$

$$\text{precision} = \frac{85}{85 + 201} \approx 0.297$$

$$\text{recall} = 1$$

$$\text{F1-score} = \frac{2}{0.297^{-1} + 1^{-1}}$$

Classifier β

	cancer	no cancer
cancer	TP: 0	FN: 85
no cancer	FP: 0	TN: 201

$$\text{accuracy} = \frac{201}{286} \approx 0.703$$

$$\text{precision} = \text{undefined}$$

$$\text{recall} = 0$$

$$\text{F1-score} = \text{undefined}$$

COVID example

	Positive	Negative
Positive	TP: 10	FN: 0
Negative	FP: 990	TN: 999,000

(Extra) Question 4/4: Classification pipeline using cross-validation (CV)

Re-arrange the following 20 steps into a correct order of a classification pipeline with 3-fold CV. There may be multiple correct orders out of 20! possible orders.

The nomenclature is as follows.

- $c(\cdot)$: a trained classifier model with some arguments inside the brackets.
- $a(\cdot)$: an accuracy score with some arguments inside the brackets.
- mm: max-min normalization technique or the corresponding result
- zs: z-score normalization technique or the corresponding result
- avg: average, the averaging operation or the corresponding result

1. Validate the trained classifier $c(\text{mm}, 1, 3)$ using fold-2 to obtain its accuracy $a(\text{mm}, 2)$.
2. Train a classifier with max-min normalization using fold-1 and fold-2. Call the resulting trained classifier as $c(\text{mm}, 1, 2)$.
3. Validate the trained classifier $c(\text{mm}, 1, 2)$ using fold-3 to obtain its accuracy $a(\text{mm}, 3)$.
4. Validate the trained classifier $c(\text{zs}, 1, 3)$ using fold-2 to obtain its accuracy $a(\text{zs}, 2)$.

5. Train a classifier with the best hyperparameter value on training set (900 data points). Call the resulting trained classifier as $c(zs)$. The training accuracy $a(zs, \text{train})$ may be obtained.
6. Train a classifier with max-min normalization using fold-2 and fold-3. Call the resulting trained classifier as $c(mm, 2, 3)$.
7. Compare the average validation accuracy: $a(zs, \text{avg})$ versus $a(mm, \text{avg})$. Say, $a(zs, \text{avg}) > a(mm, \text{avg})$. Thus, the best hyperparameter value is z-score normalization.
8. Validate the trained classifier $c(mm, 2, 3)$ using fold-1 to obtain its accuracy $a(mm, 1)$.
9. Split the dataset into training and testing sets. There 1000 data points in total, therefore 900 training data points and 100 testing data points.
10. Train a classifier with max-min normalization using fold-1 and fold-3. Call the resulting trained classifier as $c(mm, 1, 3)$.
11. Average the validation accuracy of 3 classifiers with max-min normalization. That is, $a(mm, \text{avg}) = [a(mm, 1) + a(mm, 2) + a(mm, 3)]/3$.
12. Test the trained classifier $c(zs)$ on testing set (100 data points) to obtain its the test accuracy $a(zs, \text{test})$. This $a(zs, \text{test})$ indicates the (approximate) accuracy on unseen data.
13. Validate the trained classifier $c(zs, 1, 2)$ using fold-3 to obtain its accuracy $a(zs, 3)$.
14. Train a classifier with z-score normalization using fold-2 and fold-3. Call the resulting trained classifier as $c(zs, 2, 3)$.
15. Split the training set (900 data points) into 3 folds equally. Those 3 folds are called fold-1, fold-2 and fold-3. There are 300 data points per fold.
16. Validate the trained classifier $c(zs, 2, 3)$ using fold-1 to obtain its accuracy $a(zs, 1)$.
17. Determine a set of hyperparameter values to tune. Here, there is one hyperparameter, i.e. the normalization types, namely max-min versus z-score normalizations.
18. Train a classifier with z-score normalization using fold-1 and fold-3. Call the resulting trained classifier as $c(zs, 1, 3)$.
19. Train a classifier with z-score normalization using fold-1 and fold-2. Call the resulting trained classifier as $c(zs, 1, 2)$.
20. Average the validation accuracy of 3 classifiers with z-score normalization. That is, $a(zs, \text{avg}) = [a(zs, 1) + a(zs, 2) + a(zs, 3)]/3$.